ClaimSync — CLAUDE.md

> **Single source of truth for Claude Code** Last updated: 26 Mar 2026 (EOD) — verified by Claude Code live checks Founder: Anbu, Kaaryaa GenAI Solutions Read this file before making ANY changes to ClaimSync.

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1. Product Overview

ClaimSync is a multi-tenant SaaS platform that automates downloading, classifying, and routing insurance claim files for Abu Dhabi healthcare facilities via the Shafafiya API.

Attribute	Value
Founder	Anbu, Kaaryaa GenAI Solutions
Channel Partner	Saleem — UAE operator, production sign-off authority
Target Market	Abu Dhabi healthcare facilities (Shafafiya API subscribers)
Core API	Shafafiya API (UAE DOH insurance claims ecosystem)
Cloud	Microsoft Azure — UAE North

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2. Live Versions (26 Mar 2026 EOD — verified + GitHub synced)

Component	Image	Container App / Job	Notes
Engine	`claimsync-engine:3.9`	`job-claimssync-engine`	Resubmission fix verified: 18 resub + 27 total manifest
API	`claimsync-api:2.9`	`ca-claimssync-api`	Intervals FROM/TO fix
Dashboard	`claimsync-dashboard:2.17`	`ca-claimssync-dashboard`	Interval XML viewer

All three repos synced to GitHub with detailed commit messages.

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3. Azure Resources

Resource	Value
Subscription	`ec12e27e-1ef9-46e9-8817-46a10b197381`
Tenant	`4a8ec151-3c90-4516-95c5-4c60fe32d713`
Resource Group	`rg-claimssync-uaenorth-prod` (UAE North)
Container App Env	`cae-claimssync-uae` (Consumption plan)
ACR	`crclaimssync.azurecr.io`
API FQDN	`ca-claimssync-api.whitewater-45edc27c.uaenorth.azurecontainerapps.io`
Dashboard FQDN	`ca-claimssync-dashboard.whitewater-45edc27c.uaenorth.azurecontainerapps.io`
Engine Job	`job-claimssync-engine` — cron `0 2 * * *` = 06:00 UAE
Key Vault	`kv-claimssync-uae` (RBAC mode)
DB Host	`claimssync-db.postgres.database.azure.com`
DB	`dbname=postgres` / schema=`claimssync` / user=`claimsyncadmin`
DB Password	`ClaimSync@DB2026!` (reset 25 Mar 2026)
DB KV Secret	`db-dsn` (Container App Job secret — not KV reference)
Blob Storage	`stclaimssyncuae`
Blob MF2618	`claimssync-mf2618`
Blob MF5360	`claimssync-mf5360`
Managed Identity	`id-claimssync-engine` — client_id: `8e309ea2-175e-497e-8849-7af81a36c62a`
ACS	`acs-claimssync-uae` / `acs-email-claimssync-uae`
ACS Domain	AzureManagedDomain linked ✅

Resource	Value
ACS Sender	`donotreply@3c391368-4b3b-4b02-8ae4-faf17ee6dc4a.azurecomm.net`
App Insights	`ai-claimssync-uae` (key: `6d6b4002-4e78-4628-b710-1a3585901dca`)
Log Analytics	`eb4827b3-0541-4fc4-991d-70d733e1c092`
Future Domain	`ClaimSync.Cloud` (registered — DNS not yet pointed to Azure)

4. Env Vars — API Container

Var	Source
`AZURE_CLIENT_ID`	`8e309ea2-175e-497e-8849-7af81a36c62a`
`CLAIMSSYNC_DB_DSN`	Container App secret: `db-dsn`
`CLAIMSSYNC_API_KEY`	KV secret
`CLAIMSSYNC_JWT_SECRET`	KV secret
`CLAIMSSYNC_ACS_CONNECTION_STRING`	KV secret: `acs-connection-string`
`CLAIMSSYNC_ACS_SENDER_ADDRESS`	KV secret: `acs-sender-address`
`CLAIMSSYNC_DASHBOARD_URL`	Dashboard FQDN
`APPLICATIONINSIGHTS_CONNECTION_STRING`	App Insights
`IMAGE_TAG`	Current image tag

5. Env Vars — Engine Job

Var	Source
`AZURE_CLIENT_ID`	`8e309ea2-175e-497e-8849-7af81a36c62a`
`KEYVAULT_URI` / `CLAIMSSYNC_KV_URI`	`https://kv-claimssync-uae.vault.azure.net/`
`BLOB_ACCOUNT`	`stclaimssyncuae`

Var	Source
`CLAIMSSYNC_BLOB_UPLOAD`	`1`
`CLAIMSSYNC_STORAGE_URL`	`https://stclaimssyncuae.blob.core.windows.`
`CLAIMSSYNC_TEMP_DIR`	`/tmp/claimssync/`
`CLAIMSSYNC_TENANT`	`KAARYAA-T1`
`APPLICATIONINSIGHTS_CONNECTION_STRING`	App Insights
`CLAIMSSYNC_ADHOC_FROM`	Optional — adhoc run from date
`CLAIMSSYNC_ADHOC_TO`	Optional — adhoc run to date
`CLAIMSSYNC_ADHOC_FACILITY`	Optional — single facility override

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6. Dev Directories (Windows)

Component	Path
Engine	`D:\KaaryaaDigital\Clients\ShafaAPI\ClaimsSync\ClaimSyncDocker`
API	`D:\KaaryaaDigital\Clients\ShafaAPI\ClaimsSync\ClaimSyncAPI`
Dashboard	`D:\KaaryaaDigital\Clients\ShafaAPI\ClaimsSync\ClaimSyncDashboard`

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7. Two-Environment Rule — NEVER MIX

Environment	Use For	Syntax
Windows CMD (Claude Code)	Docker builds, code edits, file ops	`dir`, `copy`, `^` continuation
Azure Cloud Shell (bash)	`az` CLI, `psql`, deployments, log queries	`ls`, `cp`, `\` continuation

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8. Admin Users

User	Role	Password
`anbu@kaaryaa.com`	Kaaryaa Super Admin	`Kaaryaa@Admin2026`
`saleem@claimsync.cloud`	Master Reseller	`ClaimSync@Saleem2026`

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9. Active Facilities

Code	Name	Tenant	Status	Blob Container
MF2618	Mediclinic	KAARYAA-T1	Active ✅	`claimssync-mf2618`
MF5360	MF5360 Facility	KAARYAA-T1	Active ✅	`claimssync-mf5360`
PF2576	(Pending)	KAARYAA-T1	Inactive ⬜	`claimssync-pf2576`

IMPORTANT: All facilities must be under `KAARYAA-T1` — engine uses `CLAIMSSYNC_TENANT=KAARYAA-T1`.

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10. Blob Storage Layout (per facility)

```
claimssync-{facility}/
  claims/                    ← H* claim XML files
  remittance/                ← RA_* extracted remittance XMLs
  resubmission/              ← RSB*/RESUB* resubmission XMLs
  search_history/            ← per-interval SOAP request + response XMLs
    search_history_request_{FAC}_{TYPE}_{N}.xml
    search_history_response_{FAC}_{TYPE}_{N}.xml
  logs/                      ← per-run audit files
    downloadlog-{FAC}-{ts}.log   ← detailed run log
    downloadlog-{FAC}-{ts}.csv   ← file manifest CSV
```

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11. Saleem PC — Local Folder Structure

```
C:\Users\USER\ClaimSync\Raggr\
  claimsync_pull_YYYY-MM-DD.log        ← central pull log (all facilities)

  mf2618\
    search_history_request_*.xml       ← from blob search_history/
    search_history_response_*.xml
    claims\
      archive\
    remittance\
      archive\
    resubmission\
      archive\
    logs\
      downloadlog-MF2618-*.log
      downloadlog-MF2618-*.csv

  mf5360\                              ← same structure
    search_history_request_*.xml
    search_history_response_*.xml
    claims\
      archive\
    remittance\
      archive\
    resubmission\
      archive\
    logs\
      downloadlog-MF5360-*.log
      downloadlog-MF5360-*.csv
```

ClaimSyncPull.py v2.0 — Task Scheduler 07:00 UAE daily. Pulls all facilities. Central log at `Raggr\`.

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12. Dashboard Routes — 18 Live + 1 Pending

All verified from codebase scan. Pure Next.js App Router.

Route	Status
`/`	Live ✅
`/facilities`	Live ✅
`/admin/login`	Live ✅
`/admin/dashboard`	Live ✅
`/admin/facilities`	Live ✅
`/admin/facilities/[code]/adhoc-run`	Live ✅
`/admin/onboarding`	Live ✅
`/admin/onboarding/[id]`	Live ✅
`/admin/resellers`	Live ✅
`/admin/revenue`	Live ✅
`/admin/users`	Live ✅
`/reseller/login`	Live ✅
`/reseller/dashboard`	Live ✅
`/reseller/facilities`	Live ✅
`/reseller/facilities/[id]`	Live ✅
`/reseller/onboard`	Live ✅
`/reseller/onboarding`	Live ✅
`/onboard/credentials/[token]`	Live ✅
`/api/claimssync/[...path]`	Live ✅ (catch-all proxy)
`/facility/*`	NOT BUILT ⬜

13. Key API Endpoints

Method	Endpoint	Purpose	Auth
GET	`/health`	Health + version	None
GET	`/docs`	Swagger UI	None
POST	`/auth/admin/login`	Admin JWT	None
POST	`/auth/reseller/login`	Reseller JWT	None
GET	`/admin/dashboard`	Platform stats	JWT Admin
GET	`/admin/facilities`	All facilities	JWT Admin
PUT	`/admin/facilities/{code}/reassign-tenant`	Reassign tenant	JWT SuperAdmin
POST	`/admin/facilities/{code}/adhoc-run`	Trigger adhoc engine run	JWT Admin
GET	`/admin/facilities/{code}/runs`	Last 10 sync runs	JWT Admin
GET	`/admin/facilities/{code}/runs/{run_id}/files`	File manifest for run	JWT Admin
GET	`/admin/facilities/{code}/runs/{run_id}/intervals`	Per-interval search history	JWT Admin
GET	`/admin/facilities/{code}/search-history/{filename}`	Raw SOAP XML from blob	JWT Admin

Method	Endpoint	Purpose	Auth
POST	`/admin/facilities/{code}/resend-token`	Revoke + reissue credential token	JWT Admin
GET	`/onboard/credentials/{token}`	Validate credential token	None
POST	`/onboard/credentials/`	Submit facility credentials	Token

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14. Database Schema

Table	Purpose
`claimssync.tenants`	Tenant records (short_code, name)
`claimssync.tenant_facilities`	Facility records per tenant
`claimssync.onboarding_requests`	Onboarding pipeline
`claimssync.credential_tokens`	Setup tokens — `valid|expired|used|revoked`
`claimssync.kaaryaa_admins`	Super admin accounts
`claimssync.resellers`	Reseller accounts (3-level)
`claimssync.sync_runs`	Engine run records per facility
`claimssync.file_manifest`	Per-file records — enriched with file_id, sender_id, receiver_id, record_count, transaction_date, transaction_timestamp
`claimssync.sync_run_log`	Detailed run logs — trigger_type: `scheduled|manual|api`

Schema: `claimssync_schema_v2.sql` + `migration_v3.sql` + auto-migration for file_manifest columns

15. Implemented Features — Complete Phase History

Phase 0 — Foundation ✅

- Azure resource provisioning
- RBAC roles — MI → KV + Blob
- DB schema v1

Phase 1 — Sync Engine ✅ (:3.9)

- `ClaimSync2.py` — multi-facility Shafafiya automation
- 2-hour interval date splitting (1000-file cap workaround)
- httpx synchronous calls (replaced curl)
- sleep(3) rate-limit between intervals
- Stale XML/CSV purged at startup
- `/tmp/claimssync/` temp dir
- DB logging — `sync_runs` + `file_manifest`
- Blob upload — claims / remittance / resubmission subfolders
- ZIP extraction — ZIPs downloaded, extracted, XMLs uploaded, ZIP deleted
- FindnMoveResubmission() — detects `<Resubmission>` tag, moves to resubmission/
- remove_attachments_from_resubmissionfiles() — strips `<Attachment>` + `<Observation>`
- **Resubmission blob upload fix** (:3.9) — scan resubmission/ after move, upload all files
- Per-interval search_history XMLs (request + response) uploaded to blob
- Per-run downloadlog-*.log + downloadlog-*.csv uploaded to blob
- file_manifest enriched — file_id, sender_id, receiver_id, record_count
- Adhoc run support — `CLAIMSSYNC_ADHOC_FROM/TO/FACILITY` env vars
- trigger_type = `manual` for adhoc, `scheduled` for cron
- Azure Container App Job — cron `0 2 * * *` (06:00 UAE)

Phase 2 — API ✅ (:2.9)

- FastAPI backend, JWT auth (admin + reseller)
- 15+ REST endpoints
- Full onboarding pipeline

- `email_service.py` — ACS email with correct EmailSendResult handling
- Azure Monitor + stdout logging (force=True fix)
- Adhoc run endpoint — triggers job via azure-mgmt-appcontainers
- Run history + file manifest endpoints
- Intervals endpoint — per-interval from/to + blob refs (FROM/TO fix :2.9)
- Search-history XML viewer endpoint
- Reassign-tenant endpoint (super-admin)

Phase 3 — Dashboard + Onboarding ✅ (:2.17)

- Next.js 14 App Router — 18+ live routes
- Admin portal — 8 routes
- Reseller portal — 6 routes
- Adhoc Run page `/admin/facilities/[code]/adhoc-run`
  - Section A: date range form + Run Now button
  - Section B: Run History table (auto-refresh 30s) + Files/Intervals tabs
  - Section C: File manifest with filter + CSV export
  - Section D: Interval Search History table
    - Per-interval FROM/TO, files found, Request/Response status
    - Side-by-side XML viewer modal (SOAP request vs response)
    - Graceful fallback for pre-:3.8 runs
- Credential setup page `/onboard/credentials/[token]`
- ResendCredentialToken.tsx — 4-state token UI
- DB schema v2 + migration_v3 (7-day token expiry)

ClaimSyncPull.py v2.0 ✅ (local — Saleem PC)

- Multi-facility: MF2618 + MF5360
- Blob prefix routing: claims/ → claims, search_history/ → root, logs/ → logs\
- Central log: `Raggr\claimsync_pull_YYYY-MM-DD.log`
- Skip-if-exists logic
- archive\ subfolders auto-created
- Task Scheduler 07:00 UAE daily

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16. Bug Fix Log

Date	Bug	Root Cause	Fix
25 Mar	Credential page "Link Not Found"	`data.valid` vs `data.status`	Normalized in `page.tsx`
25 Mar	Email no stdout logs	`configure_azure_monitor()` silenced `basicConfig`	`force=True` in `main.py`
25 Mar	Email falsely reported failed	`poller.result()` object not dict	`getattr(result, 'message_id', None)`
26 Mar	Engine DB auth failure	DB password reset but KV + CA Job secret not updated	Updated both KV + Container App Job secret
26 Mar	Engine finds 0 facilities	MF2618/MF5360 under wrong tenants (MEDICLINIC/MF5360-CLINIC vs KAARYAA-T1)	Reassigned via new API endpoint
26 Mar	trigger_type constraint violation	`sync_run_log` CHECK allows `scheduled|manual|api` not `adhoc`	Changed to `manual`
26 Mar	Blob container missing for MF5360	Never created	Created `claimssync-mf5360`
26 Mar	Interval FROM/TO showing `—`	`from_time`/`to_time` never populated in intervals endpoint	Parsed from request XML blobs in API :2.9
26 Mar	Resubmission files not in blob	Blob upload runs before `FindnMoveResubmission()` moves files	Post-move resubmission upload scan in :3.9

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17. Shafafiya API Coverage

Operation	Status	Phase
`SearchTransactions`	LIVE ✅	1

Operation	Status	Phase
`DownloadTransactionFile`	LIVE ✅	1
`GetNewTransactions`	Planned ⬜	4
`SetTransactionDownloaded`	Planned ⬜	4
`GetClaimCountReconciliation`	Planned ⬜	4
`CheckForNewPriorAuthTransactions`	Planned ⬜	4
`GetNewPriorAuthorizationTransactions`	Planned ⬜	4
`UploadTransaction`	Planned ⬜	5
`AddDRGToEClaim`	Planned ⬜	5
`GetDRGDetails`	Planned ⬜	5
`GetPersonInsuranceHistory`	Planned ⬜	5
`GetInsuranceContinuityCertificate`	Planned ⬜	6
`CancelInsuranceContinuityCertificate`	Planned ⬜	6

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18. Roadmap

Immediate — 27 Mar 2026

- ☐ Verify 06:00 UAE cron resumes for both MF2618 + MF5360 (post tenant fix)
- ✅ Resubmission fix verified — 18 resub files + 27 total manifest for MF5360 16 Mar
- ✅ PostgreSQL cost investigated — already at floor (Standard_B1ms, 32GB, 7-day backup, geo-redundant disabled). ₹3,578/mo is minimum for Azure Managed PostgreSQL UAE North. No further reduction possible without moving off managed service.
- ✅ All three repos synced to GitHub with detailed commit messages

Next Sprint

- ☐ Reporting module — per-facility, date range, CSV export
- ☐ Facility self-service portal `/facility/*`
- ☐ PF2576 activation — 3 KV secrets + DB status + blob container
- ☐ Per-interval audit logging in DB (`sync_run_intervals` table)

- [ ] Multi-tenant access control ADR-002 — JWT-scoped DB filters
- [ ] XML beautification in interval viewer (syntax highlight)
- [ ] `ClaimSync.Cloud` custom domain — DNS pointed to Azure

Phase 4 — VNet + Private Endpoints

- [ ] `cae-claimssync-uae-v2` Dedicated plan, VNet-injected
- [ ] Private Endpoints for DB + Blob + KV
- [ ] DB public access disabled

Phase 5 — Beta Launch

- [ ] Reseller self-service, billing hooks, SLA monitoring, custom domain
- [ ] Shafafiya upload APIs

Phase 6 — GenAI Intelligence

- [ ] Rejection pattern analysis, smart resubmission, NL query, anomaly detection

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19. Key Architectural Principles

1. **httpx over curl** — no race conditions, ~80s saved per run
2. **2-hour interval splitting** — Shafafiya silently caps at 1000 files per call
3. **UAE-only API** — TCP blocked outside UAE; engine must stay in Azure UAE North
4. **`/tmp/claimssync/`** — `/mnt/` is tiny ephemeral; never use for scratch
5. **Docker non-root** — `mkdir` after `useradd` in single `RUN` layer
6. **Env var hygiene** — empty values = silent `LocalINIProvider` fallback
7. **`PGPASSWORD` for psql** — `!` in password breaks bash history expansion
8. **Container App Job secret** — `db-dsn` is a manual secret in the Job, not a KV reference. Update via `az containerapp job secret set`, not just KV.
9. **All facilities must be under KAARYAA-T1** — engine uses `CLAIMSSYNC_TENANT=KAARYAA-T1`
10. **trigger_type constraint** — `sync_run_log` allows `scheduled|manual|api` only
11. **Code protection** — `.py` source never on Saleem PC; PyInstaller exe only
12. **Deployment discipline** — Saleem sign-off before every production push
13. **force=True in basicConfig** — required when `configure_azure_monitor()` also used

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20. Claude Code Ground Rules

1. ALL tasks via Claude Code prompt — code, builds, push, deploy, psql, logs
2. Never give manual CLI commands — always as a Claude Code prompt
3. Frontend browser test is the only exception — Anbu's discretion
4. Confirm current live versions before building new images
5. Bump minor version on every deploy
6. Saleem sign-off before every production deployment
7. `PGPASSWORD` for psql — never inline `!` passwords in bash
8. Windows CMD for builds; Azure Cloud Shell bash for az/psql
9. Never mix environment syntax